

Statistical Models in Computational Biology

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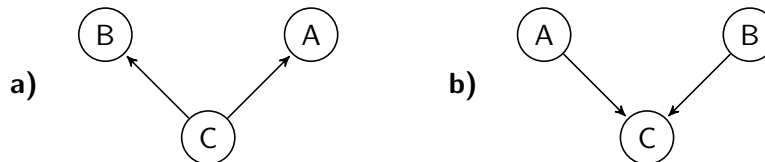
Due date: 25th Feb 2026 before 14:00

Form groups by 24th Feb and notify us via smcb@bsse.ethz.ch.

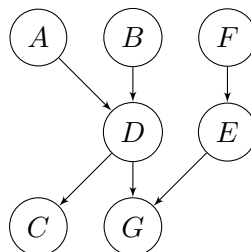
You can only submit in a group, not on your own. Submit on Moodle, not by email.

Problem 1: Conditional independence and BNs**(3 points)**

Consider the following graphical structures, corresponding to (different) Bayesian networks. For which network does the statement $A \perp B \mid C$ hold? For which does the statement $A \perp B$ hold? Prove your answers by the laws of probability.

**Problem 2: Markov blanket****(2 points)**

Consider the following graphical structure of a Bayesian network:



Determine the Markov blanket $MB(D)$ of node D and show that the conditional probability $P(D \mid A, B, C, E, F, G)$ is the same as $P(D \mid MB(D))$.

Problem 3: Learning Bayesian networks from protein data**(5 points)**

In this exercise, we will use the R package BiDAG¹ to learn Bayesian networks from protein data. Download the data [here](#). The data consists of the measurements of 11 phosphorylated proteins and phospholipids derived from primary immune system cells, subjected to both general and specific molecular interventions [2]. (*Hint: read the help files of the package and use default parameters unless otherwise stated.*)

- (a) First, run `set.seed(42)` for reproducibility. Read in the data in R. Report the number of variables n and the number of observations N . Visualize the data in a pair plot using the

¹Run `install.packages("BiDAG")` in the R console. Then load the package by running `library(BiDAG)`. You may need to install the missing packages `graph`, `RBGL` and `Rgraphviz` from Bioconductor.

`ggpairs` function in the R package `GGally`. Randomly split the data into 80% training data and 20% test data. Initialize the parameters using the function `BiDAG::scoreparameters` with the training data and the Bayesian Gaussian equivalent (BGe) score [3, 4]. (2 points)

[Note: The BGe score is a fully-decomposable marginal likelihood function $P(\mathcal{D} \mid \mathcal{G})$ for scoring Bayesian networks. The main underlying assumption is that the data is normally distributed with $\mathcal{N}(\mu, W^{-1})$. The precision matrix W follows a Wishart prior $\mathcal{W}_n(T^{-1}, \alpha_w)$, where $\alpha_w > n - 1$ is the degrees of freedom and T is the positive definite parametric matrix. The mean vector μ follows a normal prior $\mathcal{N}(\nu, \alpha_\mu W)$ with $\alpha_\mu > 0$.]

- (b) Learn a Bayesian network using the `BiDAG::iterativeMCMC` function. Plot the directed acyclic graph (DAG). Evaluate the log BGe score of the test data against the estimated DAG using the function `BiDAG::scoreagainstDAG`. (2 points)
- (c) Assume you receive new data where only phosphorylated proteins are measured but the phospholipids, PIP2 and PIP3, are not. You would like to impute PIP2 and PIP3 from (a subset of) the other variables. According to the Bayesian Network you learned in (b), which subset should you choose and why? (1 point)

References

- [1] Suter, P., Kuipers, J., Moffa, G., & Beerenwinkel, N. (2021). Bayesian structure learning and sampling of bayesian networks with the r package BiDAG. *arXiv preprint arXiv:2105.00488*.
 - [2] Sachs, K., Perez, O., Pe'er, D., Lauffenburger, D. A., & Nolan, G. P. (2005). Causal protein-signaling networks derived from multiparameter single-cell data. *Science*, 308(5721), 523-529.
 - [3] Geiger, D., & Heckerman, D. (2002). Parameter priors for directed acyclic graphical models and the characterization of several probability distributions. *The Annals of Statistics*, 30(5), 1412-1440.
 - [4] Kuipers, J., Moffa, G., & Heckerman, D. (2014). Addendum on the scoring of Gaussian directed acyclic graphical models. *The Annals of Statistics*, 42(4), 1689-1691.
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